جامعة الملك عبدالله
للعلوم والتقنية
King Abdullah University of
Science and Technology

MACHINE LEARNING

– CS229 –

Trial Exam

Total: 100 Points

Instructor: Ivan VIOLA

Student name: _____

Score: _____

Spring 2025

Problem 1: Polynomial Regression with Regularization (18 points)

Statement. We have four data points in 1D:

$$(x_1, t_1) = (0, 0), \quad (x_2, t_2) = (1,\ 0.8), \quad (x_3, t_3) = (2,\ 1.6), \quad (x_4, t_4) = (3,\ 3.7).$$

We fit the polynomial

$$y(x) \;=\; w_0 \;+\; w_1\, x \;+\; w_2\, x^2,$$

subject to a ridge (L2) penalty $\lambda = 1$:

$$\sum_{n=1}^{4} \big[y(x_n) - t_n\big]^2 \;+\; \lambda \sum_{j=0}^{2} w_j^2.$$

(a) Write the design matrix $\mathbf{\Phi}$.

(b) Formulate the ridge equation.

(c) Solve for (w_0, w_1, w_2) *numerically*, by explicitly writing and inverting the matrix.

Potentially useful inverses of matrices:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & -3 & 1 \\ -2.5 & 4 & -1.5 \\ 0.5 & -1 & 0.5 \end{pmatrix} \quad \begin{pmatrix} 5 & 6 & 14 \\ 6 & 15 & 36 \\ 14 & 36 & 99 \end{pmatrix}^{-1} = \begin{pmatrix} 0.3865 & -0.1840 & 0.0123 \\ -0.1840 & 0.6115 & -0.1963 \\ 0.0123 & -0.1963 & 0.0798 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{pmatrix}^{-1} = \begin{pmatrix} 0.95 & -1.05 & 0.25 \\ -1.05 & 2.45 & -0.75 \\ 0.25 & -0.75 & 0.25 \end{pmatrix}$$

Grading:

- *(a) Design matrix – 5 pts*
- *(b) Ridge equation form – 4 pts*
- *(c) Numeric solution with explicit matrix steps – 9 pts*

Total = 18 points

Problem 2: Multiclass Logistic Regression with Three Classes (17 points)

Statement. We have three classes $C_1,\ C_2,\ C_3$ and one 2D data point *from each class*:

$$\mathbf{x}_1 = (0,1)\ \in\ C_1, \quad \mathbf{x}_2 = (1,2)\ \in\ C_2, \quad \mathbf{x}_3 = (2,2)\ \in\ C_3.$$

Our *multiclass* logistic model has three activation functions:

$$a_1(\mathbf{x}) = w_{1,0} + w_{1,1}\,x_1 + w_{1,2}\,x_2, \quad a_2(\mathbf{x}) = w_{2,0} + w_{2,1}\,x_1 + w_{2,2}\,x_2, \quad a_3(\mathbf{x}) = w_{3,0} + w_{3,1}\,x_1 + w_{3,2}\,x_2.$$

(a) Suppose we have the parameter vectors:

$$\mathbf{w}_1 = (-1,\ 1,\ 0), \quad \mathbf{w}_2 = (-1,\ 0,\ 1), \quad \mathbf{w}_3 = (0,\ -1,\ 2).$$

Compute $a_1(\mathbf{x}_n),\ a_2(\mathbf{x}_n),\ a_3(\mathbf{x}_n)$ *and* the $\mathrm{softmax}\big(a_1, a_2, a_3\big)$ for each data point $n = 1, 2, 3$.

(b) Which class is predicted for each of the three points?

Grading:

- *(a) Softmax computations – 12 pts*
- *(b) Predicted class per point – 5 pts*

Total = 17 points

Problem 3: A Two-Layer Network with Sigmoid Hidden Units (17 points)

Statement. We have a neural network with a *single scalar input* $x \in \mathbb{R}$, two hidden units (each using a logistic sigmoid activation), and one linear output:

$$h_1 = \sigma(w_1\, x + b_1), \quad h_2 = \sigma(w_2\, x + b_2), \quad y = w_{o1}\, h_1 + w_{o2}\, h_2 + b_o,$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$.

We have two training points:

$$(x_1, t_1) = (0,\ 1.0), \quad (x_2, t_2) = (2,\ 2.0),$$

and the error function is

$$E = \tfrac{1}{2} \sum_n \big(y_n - t_n\big)^2.$$

Suppose at initialization we set *all weights* $w_1, w_2, w_{o1}, w_{o2}, b_1, b_2$ to 0 except $b_o = 1$.

(a) Write down the chain-rule expressions for $\frac{\partial E}{\partial w_j}$ and $\frac{\partial E}{\partial b_j}$ at the hidden layer.

(b) Perform one gradient descent step with a small learning rate (say $\eta = 0.1$) on (w_{o1}, w_{o2}, b_o), assuming $h_1 = h_2 = 0.5$. Show the updated values of (w_{o1}, w_{o2}, b_o).

Grading:

- *(a) Chain rule for hidden-layer parameters – 8 pts*
- *(b) Single gradient step on output layer – 9 pts*

Total = 17 points

Problem 4: A Kernel Method with $\big(x\,x' + 2\big)^2$ (14 points)

Statement. We have three one-dimensional points:

$$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3.$$

We define a polynomial kernel:

$$k(x, x') \;=\; \big(x\,x' + 2\big)^2.$$

(a) Compute the 3×3 Gram matrix $K_{ij} = k(x_i,\, x_j)$.

(b) Expand $\big(x\,x' + 2\big)^2$ to show the implied feature map $\phi(x)$.

Grading:

- *(a) Gram matrix numeric – 8 pts*
- *(b) Feature-map expansion – 6 pts*

Total: 14 points

Problem 5: Hard-Margin SVM in 2D (14 points)

Statement. We have four points in $\mathbb{R}^2$, split into two linearly separable classes:

$$\text{Class } (-1)\colon (0,0),\ (1,0), \quad \text{Class } (+1)\colon (2,2),\ (3,3).$$

We want to:

(a) By inspection, propose a line that separates these classes.

(b) Identify the support vectors (the closest points to the decision boundary in each class).

(c) Compute the margin width of your line. Is this the maximum margin?

Grading:

- *(a) Separating line – 5 pts*
- *(b) Support vectors – 6 pts*
- *(c) Margin width – 3 pts*

Total = 14 points

Problem 6: Gaussian Process (20 points)

Statement. We consider a *Gaussian Process* (GP) prior on a function $f(x)$ which has *zero mean* and uses the squared-exponential (also called RBF) kernel with amplitude α:

$$k(x, x') \;=\; \alpha \, \exp\Big[-\tfrac{(x-x')^2}{2\,\sigma^2}\Big].$$

We have three noiseless observations:

$$(x_1,\, t_1) = (0,\, 0.5), \quad (x_2,\, t_2) = (1,\, 0.7), \quad (x_3,\, t_3) = (2,\, 1.2).$$

We want the *posterior* distribution of $f(x_*)$ for a new input $x_* = 1.5$.

(a) Construct the 3×3 kernel matrix $\mathbf{K}$ for the observed inputs $\{0, 1, 2\}$.

(b) Construct the 3-vector $\mathbf{k}_*$ which consists of $k(x_*, x_n)$ for $n = 1, 2, 3$.

(c) Write down the standard GP formulas for the predictive distribution and express mean and variance at x_*.

Grading:

- *(a) Kernel matrix* $\mathbf{K}$ *– 7 pts*
- *(b) Vector* $\mathbf{k}_*$ *– 6 pts*
- *(c) Predictive distribution formulation mean and variance expressions – 7 pts*

Total = 20 points